Unit - 3(Cell Structure and Function) - 200 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

Anmol Sharma • Lesson 34 • Feb 12, 2025

Unit – 3 (Cell Structure and Function)

200 MCQs

BIOLOGYLIVE

Title: Unit – 3 (Cell Structure and Function)

1. Centrioles are associated with

(1) DNA synthesis

(2) DNA replication

(3) Spindle formation

(4) Respiration

2. Assertion : Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles together form the endomembrane system.

Reason : Functions of these organelles are coordinated so, they are considered together as a system.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false

3. Unicellular organisms are capable of

(1) Independent existence

(2) Performing essential functions of life

(3) Both (1) and (2)

(4) None of these

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4. Statement A : An elaborate network of filamentous proteinaceous structures present in the cytoplasm is called as cytoskeleton.

Statement B : Cytoskeleton consisting of microtubules, microfilaments and intermediate filaments.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

5. Statement A : The golgi apparatus principally performs the function of packaging materials.

Statement B : Golgi apparatus is the important site of formation of glycoproteins and glycolipids.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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6. Assembly of 60S and 40S subunits of ribosome gives

(1) 50S

(2) 70S

(3) 80S

(4) 100S

7. Plasma membrane is-

(1) Semipermeable

(2) Selective / differentially permeable

(3) Permeable

(4) Non-permeable

8. The centrioles in a centrosome lie ________ to each other in which each has an organisation like the______.

(1) Longitudinal, cartwheel

(2) Parallel, cartwheel

(3) Perpendicular, cartwheel

(4) Cylindrical, cartwheel

9. Which of the following membrane proteins lie on the surface of the cell?

(1) Integral proteins

(2) Peripheral proteins

(3) Both (1) and (2)

(4) Conjugate proteins

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10. Which of the following have single membrane surrounding them?

(1) Microbodies

(2) Vacuoles

(3) Golgi apparatus

(4) All of these

11. Plasmids are

(1) Naked genomic DNA

(2) Smaller DNA than genomic DNA

(3) Enveloped DNA

(4) Chromosomal DNA

12. Who was the first one to see a live cell?

(1) Anton Von Leeuwenhoek

(2) Robert Leeuwenhoek

(3) Robert Brown

(4) None of these

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13. Statement A : Ribosomes are composed of deoxy ribonucleic acid and proteins.

Statement B : Ribosomes are surrounded by membrane.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

14. Identify the membrane that surrounds the vacuole?

(1) Protoplast

(2) Tonoplast

(3) Cell membrane

(4) None of these

15. Small particles projecting from the inner membrane of mitochondria are known as

(1) Microsome

(2) Macrosome

(3) F0-F1particles

(4) Lysosomes

16. The stroma of chloroplast contains enzymes required for synthesis of

(1) Carbohydrates

(2) Proteins

(3) Fats

(4) Both 1 and 2

17. Which component of plasma membrane increase the fluidity?

(1) Phospholipids

(2) Cholesterol

(3) Carbohydrates

(4) Proteins

18. Which one of the following is genetically active and contain loosely packed DNA?

(1) Heterochromatin

(2) Euchromatin

(3) Nucleolus

(4) Mitochondria

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19. Spindle fibres attach on to____________.

(1) Telomere of the chromosome

(2) Kinetochore of the chromosome

(3) Centromere of the chromosome

(4) Kinetosome of the chromosome

20. Choose the incorrect statement.

(1) Human red blood cells are about 7.0 μm in diameter

(2) Bacteria are 3-5 μm in lenght

(3) The largest cell is the egg of an ostrich

(4) Mycoplasma is the largest cell

21. Nuclear DNA exists as a complex of proteins called –

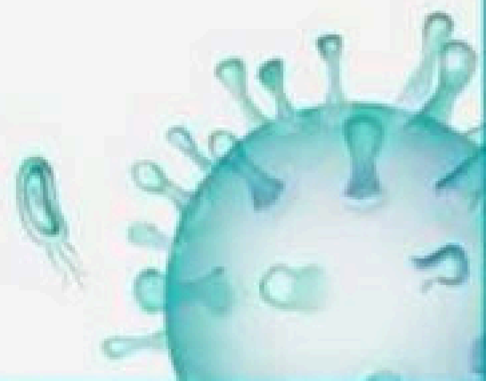
that condenses in to – during –

(1) Chromatids, Chromosomes, Cell division

(2) Chromosomes, Chromatin, Interphase

(3) Chromatin, Chromosome, Cell Division

(4) Chromatin, Chromosome, Interphase



22. Non-membrane bound organelles present in animal cell that helps in cell division.

(1) Nucleus

(2) Ribosomes

(3) Centriole

(4) Golgi apparatus

23. Statement A : Singer and Nicolson in 1927 proposed widely accepted asfluid mosaic model of plasma membrane.

Statement B : According to this, the quasifluid nature of lipid enables lateral movement of proteins within the overall bilayer.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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24. Choose the right sequence/route of the secretory product?

(1) RER → GB → Lysosome →Nuclear membrane →

Plasma membrane

(2) ER → Vesicles → Cis region of GB → Trans region of

GB → Plasma membrane

(3) ER → Vesicles → Trans region of GB → C is region of

GB → Vesicles → Plasma membrane

(4) Lysosome → ER → GB → Vesicles→ Cell membrane

25. Axoneme having 9+2 doublet microtubule arrangement is found in

(1) Centriole

(2) Flagella

(3) Cilia

(4) Cilia and Flagella

26. Plants cell differs from animals cell in having -

(1) Cell membrane, plastid and cell wall

(2) Cell wall, plastid, centriole

(3) Chloroplast, plastid and cell wall

(4) Large vacuole, plastid and cell wall

27. Statement A : All living organisms are composed of cell or cells.

Statement B : Unicellular organisms are composed of many cells.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

28. Select the correct statement:

(1) The chloroplast contains cristae and carotenoid pigments.

(2) Leucoplast contains fat soluble carotenoid pigments like carotene, xanthophylls.

(3) Plastid is easily observed under microscope.

(4) Chloroplast is a single membrane bound organelle.

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29. Statement A : Human red blood cells are about 7.0 μm in diameter.

Statement B : The shape of the cell may vary with the function they perform.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

30. Statement A : Passive transport does not require energy.

Statement B : Active transport also does not require energy.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

31. Which one of the following has majority of the chloroplasts?

(1) Mesophyll cells of the leaves

(2) Mesophyll cells of the stem

(3) Mesophyll cells of the roots

(4) Mesophyll cells of the flowers

32. Which cell organelle connects nuclear envelope with cell membrane?

(1) Nucleus itself

(2) Golgi body

(3) Endoplasmic reticulum

(4) Mitochondria

33. Cellular organelles with membranes are

(1) Lysosomes, Golgi apparatus and mitochondria

(2) Nuclei, ribosomes and endoplasmic reticulum

(3) Chromosomes, ribosomes and endoplasmic reticulum

(4) Endoplasmic reticulum, ribosome and nuclei

34. Statement A : Anton Von Leeuwenhoek first saw and described a live cell.

Statement B : Robert Brown later discovered the nucleolus.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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35. Microbodies are present in

(1) Plants

(2) Animals

(3) Both 1 and 2

(4) None of the above

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36. Which one of the following is formed by packaging in golgi apparatus?

(1) Lysosomes

(2) Endoplasmic reticulum

(3) Both (1) and (2)

(4) Mitochondria

37. Which of the following is a type of plastid?

(1) Chloroplast

(2) Chromoplast

(3) Leucoplast

(4) All of these

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38. Statement A : In prokaryotes the genetic material is basically enveloped by a nuclear membrane.

Statement B : Many bacteria have small circular DNA outside the genomic DNA, called Plasmids.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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39. Which of the following sequence is correct?

(1) Basal body → Cilium/flagellum → Centriole

(2) Cilium/flagellum → Basal body → Centriole

(3) Basal body → Centriole → Flagellum/ Cilium

(4) Centriole → Basal body → Cilium/ Flagellum

40. What is the length of mycoplasm cell?

(1) 3 mm

(2) 0.3 μm

(3) 3 μm

(4) 30 μm

41. Biochemical investigation revealed that the cell membranes also possess

(1) Lipids and Carbohydrate

(2) Carbohydrate only

(3) Protein only

(4) Protein and Carbohydrate

42. Statement A : In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm.

Statement B : The cytoplasm is the main arena of cellular activities in only plant cells but not in animal cells.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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43. Which of the following statements regarding cilia is correct?

(1) The axoneme possesses only one microtubul

(2) The axoneme possesses a number of microtubules running parallel to the long axis

(3) The axoneme possesses a number of microtubules running perepependicular to the long axis

(4) It varies depending on the organisms

44. Microtubules, motor proteins, and actin filaments are all part of the

(1) Mechanism of photosynthesis that occurs in chloroplasts.

(2) Rough ER in prokaryotic cells

(3) Cytoskeleton of eukaryotic cells.

(4) Process that moves small molecules across cell membranes.

45. Chlorophyll pigments are present in

(1) Thylakoid

(2) Stroma

(3) Outer membrane

(4) Inner membrane

46. Bacterial flagellum has

(1) Filament

(2) Hook

(3) Basal body

(4) All of these

47. The ability of proteins to move within the membrane is measured as

(1) Plasticity

(2) Flexibility

(3) Fluidity

(4) None of these

48. Statement A : Matthias Schleiden, a German Zoologist observed that all plants are composed of different kinds of cells

Statement B : Theodore Schwann, a British Botanist studied different types of animal cells and reported plasma membrane

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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49. Assertion : F0-F1 particles are present in the inner mitochondrial membrane.

Reason : An electron gradient formed on the inner mitochondrial membrane forms ATP.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false

50. What is true for a prokaryotic cell?

i. Prokaryotic cell is usually 0.1-2.0 micron.

ii. They are smaller in size

iii. Greatly vary in shape and size

iv. Multiply much rapidly than eukaryotes

(1) i, ii only

(2) i, ii, iii only

(3) i, iii, iv only

(4) All of these

51. Chromatophores take part in :

(1) Respiration

(2) Photosynthesis

(3) Growth

(4) Movement

52. Nucleolus is

(1) Rounded structure found in cytoplasm near nucleus.

(2) Rounded structure inside nucleus and having rRNA.

(3) Rod-shaped structure in cytoplasm near the nucleus.

(4) None of the above.

53. In flagella, there are radial spokes.

(1) 6

(2) 7

(3) 1

(4) 9

54. Kinetochore is

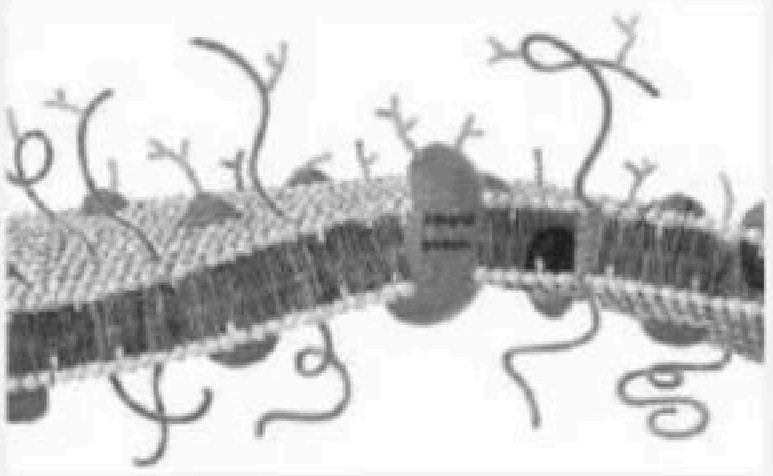
(1) Centromere

(2) Protein discs at primary constriction

(3) Protein discs at secondary constriction

(4) DNA at centromere

55. Find out correct statement about the given diagram. AScientists, B-Year, C- Structure.



(1) A - Singer and Nicolson, B - 1972, C – Nuclear membrane

(2) A - Singer and Nicolson, B - 1972, C - cell membrane

(3) A - Robert brown, B - 1831, C - Nucleus

(4) A - Matthias Schleiden, B - 1838, C - Bacteria

56. Identify the most appropriate statement about the centrioles in a centrosome

a. Each of the peripheral fibril is a triplet.

b. The adjacent triplets are linked.

(1) a is correct

(2) b is correct

(3) Both are correct

(4) Both are incorrect

57. What are flat membranous tubules connecting the different grana called?

(1) Thylakoid

(2) Granum

(3) Stroma

(4) Stroma lamellae

58. Which one of the following is ground substance of chloroplast?

(1) Stroma

(2) Granum

(3) Thylakoid

(4) Stroma lamella

59. Which among the following are small bristles like fibres sprouting out of the cell?

(1) Pili

(2) Fimbriae

(3) Flagella

(4) Mitochondria

60. Lysosome are generally found in

(1) Animal cells

(2) Plant cells

(3) both (1) and (2)

(4) Bacterial cells

61. The function of cytoskeleton is

(1) To facilitate intracellular transport

(2) To maintain cell shape and structure

(3) To provide support to the organelles

(4) All of the above

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62. The following statements tells us about a particular structure, identify it

i. It is semi-permeable in nature and interacts with the outside world.

ii. It is similar structurally to that of the eukaryotes.

(1) Capsule (2) Cell wall

(3) Cell membrane (4) Cytoplasm

63. A common characteristic feature of plant sieve tube cells and most mammalian erythrocytes is

(1) Absence of nucleus and haemoglobin

(2) Absence of nucleus and cell wall

(3) Absence of mitochondria

(4) Absence of nucleus

64. Assertion : Plastids are semiautonomous organelles.

Reason : Leucoplasts, Chromoplasts, Chloroplasts are plastids.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false

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65. The axoneme usually has ___pair of centrally located microtubules and ______pairs of doublets of radially arranged peripheral microtubules,

(1) Nine, one

(2) One, nine

(3) Nine, zero

(4) Zero, nine

66. Actively functional nucleus shows-

(1) Large nucleolus, diffused chromatin and no nuclear pores

(2) Large nucleolous, diffused chromatin and more nuclear pores

(3) Large nucleolus, compact chromatin and many pores

(4) No nucleus, diffused chromatin and small nuclear pores

67. Golgi apparatus is often seen associated with

(1) Mitochondria

(2) ER

(3) Lysosome

(4) Nucleus

68. Which of the following is incorrect matching?

(1) Round and biconcave – WBC

(2) Long and narrow – Columnar cell

(3) Elongated – Tracheid

(4) Long and branched – Nerve cells

69. The DNA present in chloroplast is:

(1) Linear, double stranded

(2) Circular, double stranded

(3) Linear, single stranded

(4) Circular, single stranded

70. Which of the following are not hydrolytic enzyme?

(1) Nucleases

(2) Proteases

(3) Carbohydrases

(4) Ligases

71. In which part of the nucleus does ribosome is produced

(1) Within nuclear pore

(2) In the chromatin

(3) In the area of the nucleolus

(4) Ribosome production doesnot occur in the nuclei

72. In bacterial cilia, the arrangement of microtubules is

____

(1) 9 + 1

(2) 9 + 0

(3) 9 + 2

(4) 0

73. Lipids are arranged within the membrane with

(1) Polar heads towards innerside and the hydrophobic

tails towards outerside

(2) Both heads and tails towards outerside

(3) Polar heads towards outerside and tails towards inside

(4) Both heads and tails towards innerside

74. Cell wall in plants consists of

(1) Lignin, hemicellulose, pectin and lipid

(2) Cellulose, hemicellulose, pectins and proteins.

(3) Lignin, hemicelluloses, protein and lipid

(4) Hemicellulose, cellulose, tubulin and lignin

75. Select the wrong statement:

(1) Bacteria exhibits 4 basic shapes

(2) Pili and fimbriae are mainly involved in motility of bacterial cells

(3) Cyanobacteria lack flagellated cells

(4) Mycoplasma is a wall less microorganism

76. The ________ of the cell directs its overall activites as well as houses its genetic material

(1) Nucleolus

(2) Endoplasmic reticulum

(3) Nucleus

(4) Centromere

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77. Which one of the following is absent in animal cell?

(1) Cellwall only

(2) Plastids and cell wall

(3) Plastids only

(4) Cytoplasm

78. Statement A : Chloroplast contains double circular DNA molecule and ribosomes.

Statement B : The ribosomes of the chloroplasts are larger than the cytoplasmic ribosomes.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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79. Which of the following is not a feature of the plasmids?

(1) Independent replication

(2) Circular structure

(3) Transferable

(4) Single stranded

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80. A few cells and associated entities are listed. Which of them represents the correct ascending order of the size relative to each other

(1) Mitochondrion < Paramecium < Human < erythorocyte < E. coli

(2) Virus < Protein < Paramecium < Chloroplast

(3) Protein < Virus < Mitochondrion < Chloroplast < human RBC

(4) Nucleus < Protein < Paramecium < Chloroplast

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81. The colour of leucoplast is

(1) Yellow

(2) Green

(3) Orange

(4) Colourless

82. Bacterial cell envelope is composed of

(1) Outermost cell wall followed by glycocalyx and plasma membrane

(2) Outermost plasma membrane followed by cell wall and glycocalyx

(3) Outermost glycocalyx followed by cell wall and plasma membrane

(4) Plasma membrane and glycocalyx only

83. Of the two membranes of the chloroplast, the inner chloroplast membrane is _______

(1) Totally impermeable

(2) Relatively more permeable

(3) Relatively less permeable.

(4) None are correct

84. What is the ratio of microtubules in centrioles and flagella?

(1) 27:18

(2) 20:27

(3) 1:1

(4) 20:9

85. Statement A : Chromosome with two equal arms is called submetacentric chromosome.

Statement B : Chromosome with one shorter arm and one longer arm is called metacentric chromosome.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

86. ER divides the_____ space into _____ distinct compartments

(1) Intracellular, three

(2) Extracellular, two

(3) Intracellular, two

(4) Extracellular, three

87. Which of the following organelle is present only in animal cells?

(1) Membrane -bound - Mitochondria

(2) Non-membrane -bound - Centriole

(3) Membrane -bound - Vacuole

(4) Membrane -bound - Centriole

88. Centrosome is an organelle usually containing

(1) Two cylindrical structures called centrioles

(2) Two spherical structures called centrioles.

(3) One cylindrical structure called centriole

(4) One spherical structures called centrioles.

89. Statement A : Na+/K+ Pump is an example of Passive transporter.

Statement B : Movement of alcohol by diffusion is called osmosis.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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90. Select the correct answer.

A. Prokaryotes have something unique in the form of inclusions

B. Inclusions are essentially infoldings of cell membrane

(1) A is correct

(2) B is correct

(3) Both are correct

(4) Both are incorrect

91. Select the wrong statements from the following:

(1) Both chloroplast and mitochondria contain an inner and outer membrane

(2) Both chloroplast and mitochondria have an internal compartment,always called as stroma

(3) The chloroplasts are generally much larger than mitochondria

(4) Both chloroplast and mitochondria contain DNA and 70 S ribosomes

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92. Statement A : In Amoeba the contractile vacuole is important for osmoregulation and excretion.

Statement B : In many cells, as in protists, contractile vacuoles are formed by engulfing the food particles.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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93. Select the correct answer

(A)The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins

(B) The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates only

(1) A is correct only

(2) B is correct only

(3) Both are correct

(4) None are correct

94. Lipid bilayer is

(1) Hydrophilic

(2) Hydrophobic

(3) Hydrophilic and Hydrophobic

(4) Depends on the surrounding medium

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95. Statement A : The fimbriae are elongated tubular structures made of a special protein.

Statement B : The pili are small bristle like fibres sprouting out of the cell.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

96. Cis and trans face of golgi body are ____ and ____ respectively.

(1) Convex-forming, Concave -maturing

(2) Concave-maturing, Convex-forming

(3) Convex-maturing, Convex -forming

(4) Concave-forming-, Concave maturing

97. A cell, which is very active in the synthesis and secretion of proteins, would be expected to have-

(1) Equal amount of RER and SER

(2) More SER than RER

(3) More RER than SER

(4) It is always variable

98. For the study of structure of nucleus, the best cell is

(1) Cell in any stage of mitosis

(2) Cell in the late prophase

(3) Cell in the sphase

(4) Cell in the interphase

99. What is the size of a typical eukaryotic cell?

(1) 1-2 μm

(2) 10-20 μm

(3) 10-20 mm

(4) 10-20 cm

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100. Identify the round and oval plant cell and amoeboid human cell respectively?

(1) Mesophyll cells, RBCs

(2) Mesophyll cells, WBCs

(3) Columnar cells, Mesophyll cells

(4) Mesophyll cells, Nerve cell

101. Which of the following is correct about human Haemoglobin (Hb)?

(1) Made up to 2-alpha and 2-beta subunits

(2) Present in erythrocytes

(3) Use to carry O_2 and CO_2

(4) All of these

102. The percentage of carbon in earths crust is

(1) 65%

(2) 46.6%

(3) 18.5%

(4) 0.03%

103. Which of the following is an example of Phospholipid?

(1) Prostaglandins

(2) Lecithin

(3) Bile acids

(4) Glycerol

104. With reference to enzymes, which one of the following statement is true?

(1) Apoenzyme = Holoenzyme + Coenzyme

(2) Holoenzyme = Apoenzyme + Coenzyme

(3) Coenzyme = Apoenzyme + Holoenzyme

(4) Holoenzyme = Coenzyme - Apoenzyme

105. Statement A : When ice melting to water is an example of chemical reaction.

Statement B : Hydrolysis of starch into glucose is a physical process.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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106. An amino acid under certain conditions has both positive and negative charges simultaneously in the same molecule. Such a form of amino acid is called

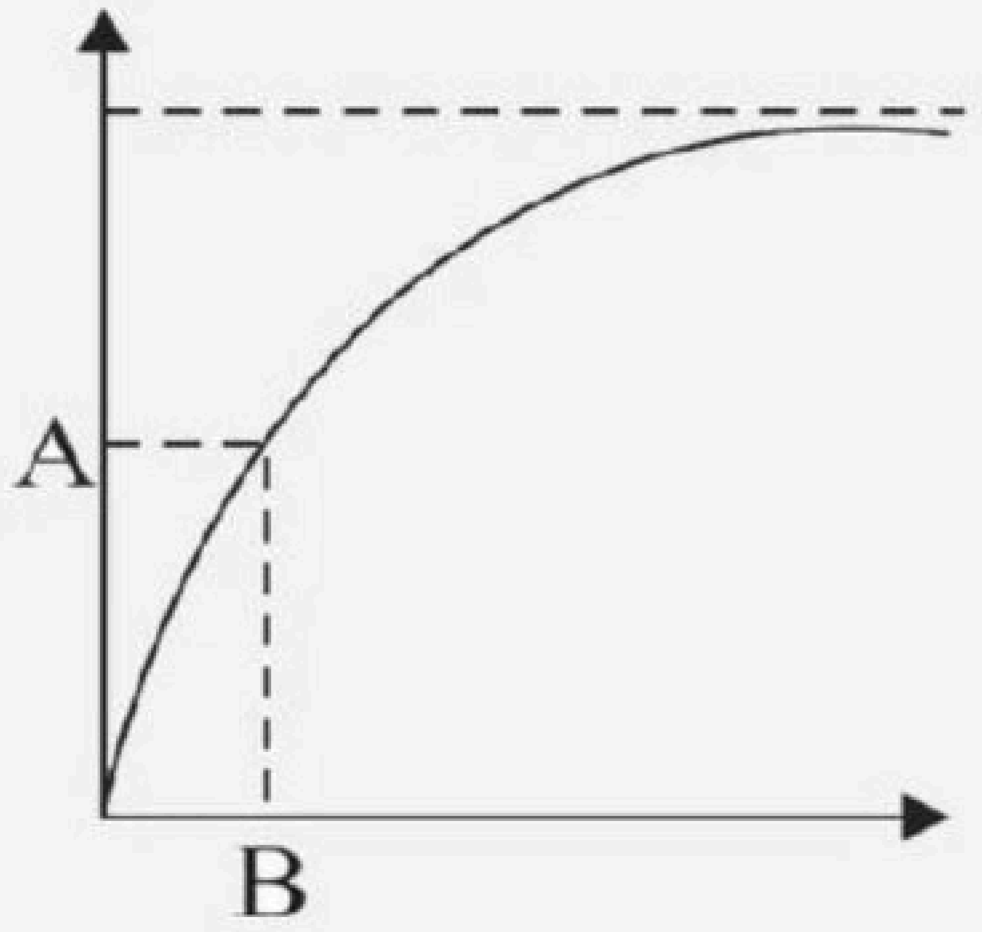
(1) Acidic form

(2) Basic form

(3) Aromatic form

(4) Zwitter ionic form

107. Identify A and B in the given diagram?



(1) A - Velocity of reaction, B - Concentration of substrate

(2) A - Velocity of reaction, B - Temperature

(3) A - Velocity of reaction, B - Km

(4) A - Temperature, B - Velocity of reaction

108. Triglycerides consist of a glycerol molecule bonded to how many numbers of fatty acids?

(1) 2

(2) 3

(3) 4

(4) 1

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109. Assertion : A coenzyme or metal ion tightly bound to enzyme protein is called prosthetic group.

Reason : A complete catalytically active enzyme together with its bound prosthetic group is called apoenzyme

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

110. Similarity in DNA and RNA is that

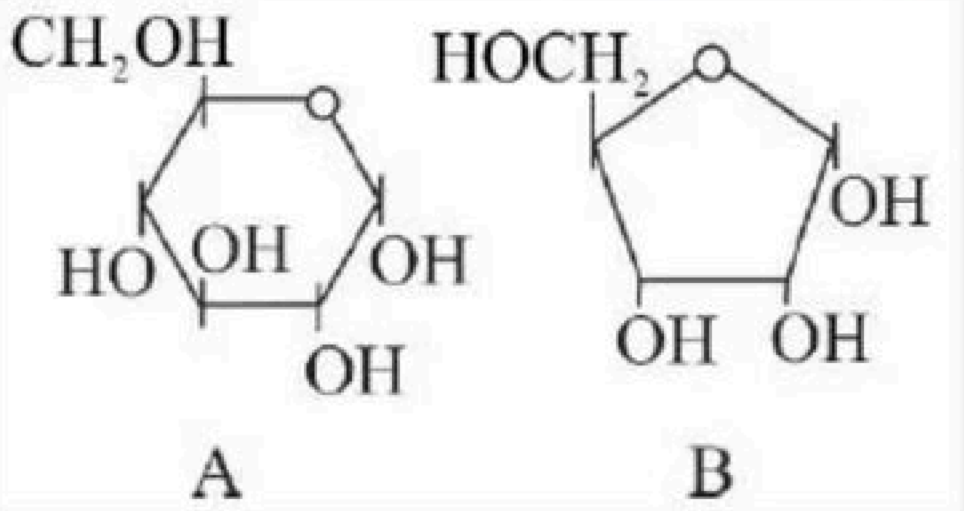
(1) Both are polymers of nucleotides

(2) Both have similar pyrimidines

(3) Both have similar sugar

(4) Both are non genetic material in man

111. Identify A and B in the given diagram?



(1) A - Fructose, B - Glucose

(2) A - Glucose, B - Ribose

(3) A - Ribose, B - Glucose

(4) A - Galactose, B - Ribose

112. A non-proteinaceous enzyme is :-

(1) Ligase

(2) Deoxyribonuclease

(3) Lysozyme

(4) Ribozyme

113. An acid soluble compound formed by phosphorylation of nucleoside is called

(1) Nitrogen base

(2) Adenine

(3) Sugar phosphate

(4) Nucleotide

114. Flow of metabolites through metabolic pathway has___

(1) Indefinite rate and direction

(2) Definite rate and direction

(3) Definite rate and no direction

(4) No fix rate but direction

115. Regarding catalytic cycle of an enzyme action, select the correct sequential steps :

I. Substrate enzyme complex formation.

II. Free enzyme ready to bind with another substrate.

III. Release of products.

IV. Chemical bonds of the substrate broken.

V. Substrate binding to active site.

Choose the correct answer from the options given below :

(1) V - I - IV - III – II　　(2) I - V - II - IV - III

(3) II - I - III - IV – V　　(4) V - IV - III - II - I

116. Nucleic acids exhibit a wide variety of ________ structures.

(1) Secondary

(2) Primary

(3) Tertiary

(4) Quarternary

117. Cellulose consists of which type of unit?

(1) Galactose

(2) Glucose

(3) Mannose

(4) Ribose

118. A homopolymer has only one type of building block called monomer repeated 'n' number of times. A heteropolymer has more than one type of monomer. Proteins are heteropolymers usually made of

(1) 20 types of monomers

(2) 40 types of monomers

(3) 30 types of monomers

(4) only one type of monomer

Title: Unit – 3 (Cell Structure and Function)

119. Identify the mismatch

(1) Alkaloids – Morphine, Codeine

(2) Lectins –. Abrin and Ricin

(3) Pigments – Carotenoids, Anthocyanin

(4) Terpenoides – Monoterpenes, Diterpenes

Title: Unit – 3 (Cell Structure and Function)

120. Select the correct matching for component percentage in living cell:

(1) Water – 7 to 59 %

(2) Proteins – 1 to 15%

(3) Carbohydrate – 12%

(4) Nucleic Acid – 5 to 7%

Title: Unit – 3 (Cell Structure and Function)

121. Assertion : Coenzyme is a non-protein group without which certain enzymes are inactive or incomplete.

Reason : Coenzymes not only provide a point of attachment of the chemical group being transformed, but also influence the properties of the group.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

122. Half Vmax is indicated by

(1) K_M

(2) K_m

(3) k_M

(4) k_m

123. DNA and RNA are

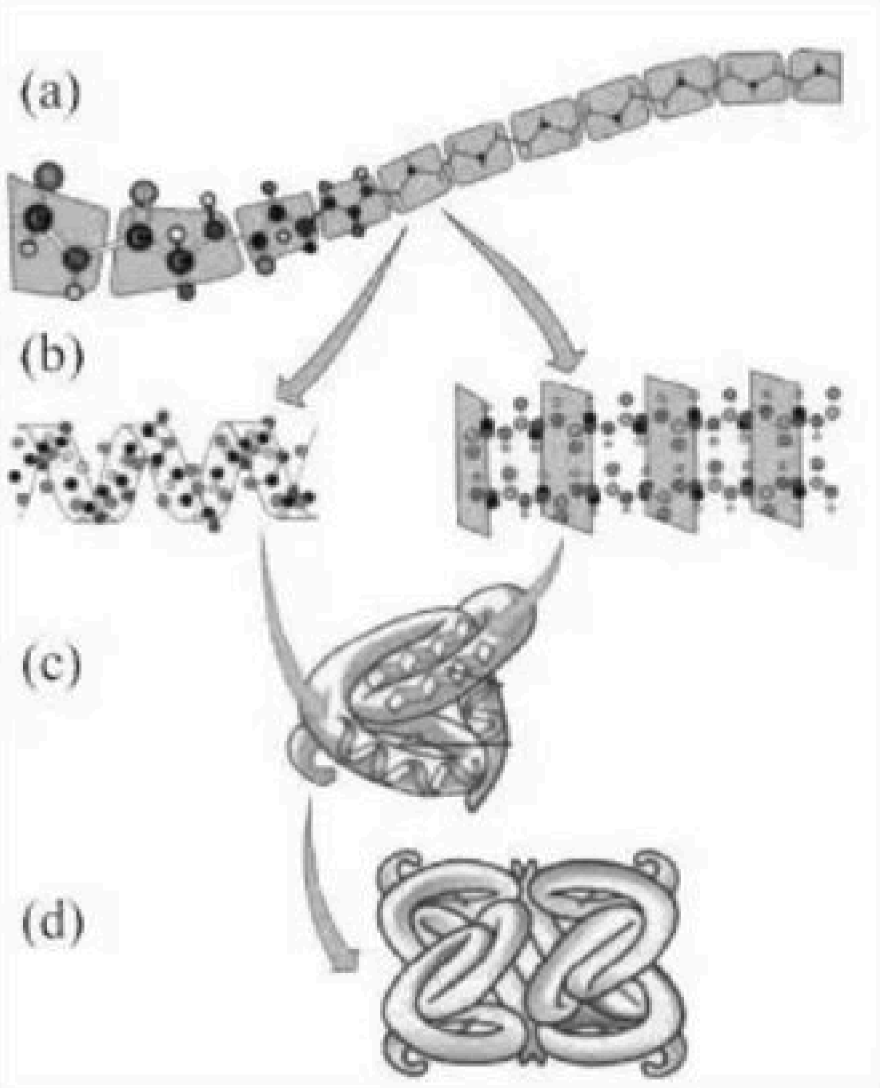
(1) Polypeptides

(2) Polynucleotides

(3) Polysaccharides

(4) None of them

124. Following diagram shows different levels of protein structure. Identify the labels a to d in the diagram?



(1) a - Quaternary, b - Tertiary, c - Primary, d - Secondary

(2) a - Secondary, b - Primary, c - Quaternary, d - Tertiary

(3) a - Primary, b - Secondary, c - Tertiary, d - Quaternary

(4) a - Secondary, b - Quaternary, c - Tertiary, d - Primary

125. Enzymes get denatured?

(1) By destroying tertiary structure

(2) By destroying primary structure

(3) By destroying secondary structure

(4) All of them

126. Enzymes catalysing the linking together of 2 compounds, e.g., enzymes which catalyse joining of CO, C-S, C-N, P-O etc. bonds

(1) Hydrolases

(2) Lyase

(3) Ligase

(4) Both (2) and (3)

127. The type of Polysaccharide present in a cotton fibre

(1) Cellulose

(2) Starch

(3) Glycogen

(4) Insulin

128. Which pyrimidine base is found in RNA only?

(1) Thymine

(2) Uracil

(3) Cytosine

(4) 2 and 3

129. Select the option which is not correct

(1) Malonate is a competitive inhibitor of succinic dehydrogenase

(2) Addition of lot of succinate does not reverse the inhibition of succinic dehydrogenase by malonate

(3) Increase in substrate concentration, increases the velocity of enzymatic reaction at first then it reaches maximum and further increment does not occur.

(4) All are correct

130. Which of the following is not a drug?

(1) Morphine

(2) Codeine

(3) Carotenoids and anthocyanin

(4) Both 1 and 2

131. An organic substance bound to an enzyme and essential for its activity is called

(1) Isoenzyme

(2) Coenzyme

(3) Holoenzyme

(4) Apoenzyme

132. Refer to the given reactions.

I. Adenine+ X + Y → Adenylic acid

Choose the correct option for X and Y.

(1) X - Phosphate group, Y - Sugar molecule

(2) X - Nitrogen base, Y - Sugar molecule

(3) X - Sugar molecule, Y - Nitrogen base

(4) X - Sugar molecule, Y - Phosphate group

Title: Unit – 3 (Cell Structure and Function)

133. Which one is most abundant protein in the animal world?

(1) Trypsin

(2) Haemoglobin

(3) Collagen

(4) Insulin

134. Which of the following is an example of sterol in animal tissue?

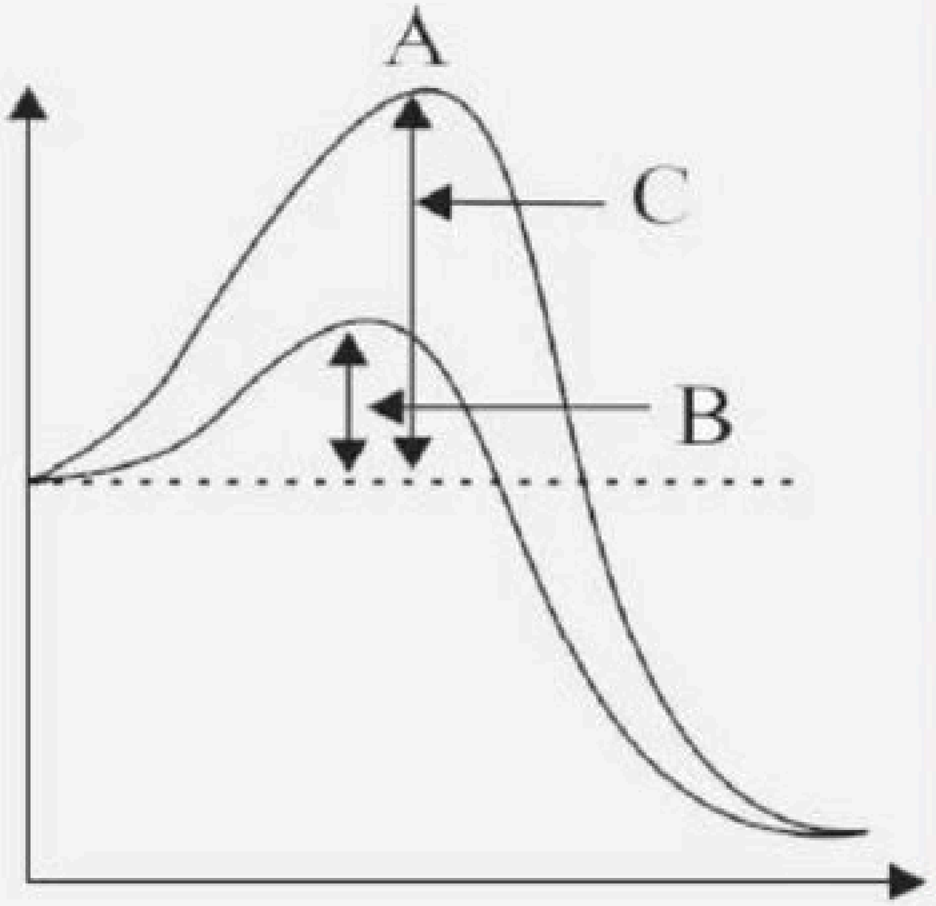
(1) Cholesterol

(2) Campesterol

(3) Stigmasterol

(4) More than one options are correct

135. Identify A, B and C in the given diagram?



(1) A - Activation energy without enzyme, B - Transition state, C - Activation energy with enzyme

(2) A - Transition state, B - Activation energy without enzyme, C - Potential energy.

(3) A - Activation energy with enzyme, B - Transition state, C - Activation energy without enzyme

(4) A - Transition state, B - Activation energy with enzyme, C - Activation energy without enzyme.

136. Assertion : Amino acids are monomers of nucleic acids

Reason : Protein amino acids have an unlimited variety.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

137. Iodine test can detect the presence of

(1) Starch

(2) Cellulose

(3) Both (1) and (2)

(4) Chitin

138. What is/are the functions of proteins?

(1) Transport nutrients across cell membrane

(2) Fights infectious organisms

(3) Acts as respiratory substrates

(4) More than one options are correct

139. Select the incorrect statement from the following:

(1) Starch forms helical secondary structures.

(2) Starch does not give iodine test.

(3) Cellulose is a homopolysaccharide.

(4) None of these

140. The inhibitor, which binds to the enzyme at site other than the active site and does not resemble the substrate in structure is called

(1) Activator

(2) Substrate analogue

(3) Competitive inhibitor

(4) Non-competitive inhibitor

Title: Unit – 3 (Cell Structure and Function)

141. Which among the following are the characteristic features of lipids?

a. Lipids are important for cell membrane structure

b. They also functions as chemical messengers in human body

c. Lipids are also found in the blood cells and in the brain

d. Lipids are composed of carbon, hydrogen and oxygen in common and sometimes also contain phosphorous, sulphur, etc

(1) a,c
(2) a,c,d
(3) a,b,c,d
(4) b,c,d

142. How many types of cofactor can be identified?

(1) 1

(2) 2

(3) 3

(4) 4

143. Hydrolysis of sucrose yields :

(1) Glucose and fructose

(2) Glucose and maltose

(3) Glucose and galactose

(4) Only glucose

Title: Unit – 3 (Cell Structure and Function)

144. Match the columns I (Component) with II (% of the total cellular mass) and choose the correct option.

Column I	Column II
A Nucleic acids	P 2
B Carbohydrates	Q 5-7
C Proteins	R 3
D Lipids	S 10-15

(1) A-R, B-S, C-P, D-Q
(2) A-Q, B-R, C-S, D-P
(3) A-Q, B-P, C-R, D-S
(4) A-S, B-P, C-Q, D-R

145. The catalytic efficiency of two different enzymes can be compared by the

(1) The Km value

(2) The optimum value

(3) Molecular size of enzyme

(4) The temperature optimum value

Title: Unit – 3 (Cell Structure and Function)

146. In the primary structure of protein

(1) Left end represents → 1st amino acid (C-terminal amino acid)

(2) Right end represents → Last amino acid (N terminal amino acid)

(3) Left end represents → 1st amino acid (N-terminal amino acid)

(4) Right end represents → 1st amino acid (C-terminal amino acid)

Title: Unit – 3 (Cell Structure and Function)

147. Statement A : Arachidonic acid has 20 carbon atoms excluding the carboxyl carbon

Statement B : Arachidonic acid is a saturated fatty acid.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

148. Destructive effect of high temperature on enzyme action is due to

(1) Inactivation of protein

(2) Denaturation of protein

(3) Acceleration of reverse reaction

(4) Formation of protein

149. Select the correct option, which represents the homopolysaccharides made up of glucose monomers.

(1) Sucrose, lactose, chitin

(2) Starch, glycogen, cellulose

(3) Starch, inulin, peptidoglycan

(4) Glucose, Starch, Cellulose

150. For its activity, carboxypeptidase requires:

(1) Zinc

(2) Iron

(3) Niacin

(4) Copper

151. Enzymes are divided into __classes each with ____subclasses and named accordingly by a _____digit number

(1) 4, 6. 13

(2) 6, 4, 13

(3) 6 , 4-13, 4

(4) Both (1) and (3)

152. Match the stages of meiosis in Column I to their characteristic features in Column II and select the correct option using the codes given below.

Column I		Column II	
A	Pachytene	1	Pairing of homologous chromosomes
B	Metaphase I	2	Terminalisation of chaismata
C	Diakinesis	3	Crossing over takes place
D	Zygotene	4	Chomosomes align at equatorial plate

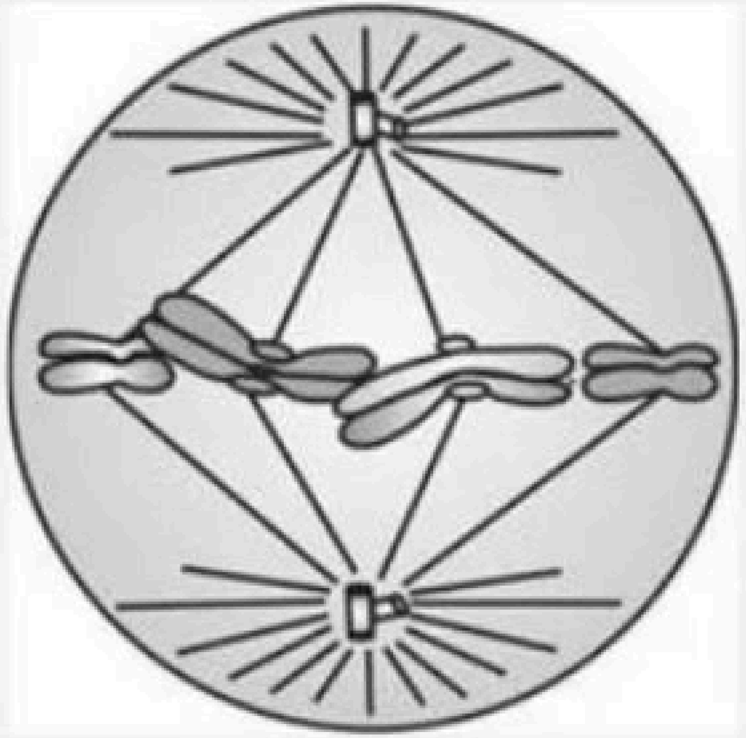
(1) A-3, B-4, C-2, D-1

(2) A-4, B-1, C-2, D-3

(3) A-4, B-3, C-1, D-2

(4) A-4, B-2, C-1, D-3

153. Given diagram represents metaphase of mitosis, chromosomes are arranged at the equator which are made up of -



(1) Two sister chromatids

(2) Two non-sister chromatids

(3) Two homologous chromosomes

(4) Two non-homologous chromosomes

154. Statement A : Prophase which is the first stage of Cytokinesis of mitosis follows the G_1 and G_0 phases of interphase.

Statement B : Prophase is marked by the completion of condensation of chromosomal material.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

Title: Unit – 3 (Cell Structure and Function)

155. Statement A : A typical eukaryotic cell divide once in approximately every 22 hours.

Statement B : Yeast for example, can progress through the cell cycle in only about 30 minutes.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

Title: Unit – 3 (Cell Structure and Function)

156. Which does not occurs in prophase:

(1) Formation of chromatin

(2) Movement of centrioles

(3) Appearance of sister chromatids

(4) Compaction of chromosome

157. Identify the meiotic stage in which the homologous chromosomes separate while the sister chromatids remain associated at their centromeres.

(1) Metaphase I

(2) Metaphase II

(3) Anaphase I

(4) Anaphase II

158. As compared to meiosis, in mitosis

(1) Homologous chromosomes pair

(2) Homologous chromosomes separate

(3) Anaphase stage is absent

(4) Prophase is shorter

159. Match the columns I and II and choose the correct option.

Column I	Column II
A Synaptonemal complex formation	P Pachytene
B Appears as tetrads	Q Leptotene
C Chromosomes become gradually visible under the light microscope	R Zygotene
D X-shaped structures	S Chiasmata

(1) A-R, B-P, C-Q, D-S
(2) A-Q, B-P, C-R, D-S
(3) A-S, B-P, C-R, D-Q
(4) A-P, B-Q, C-R, D-S

160. The spindle fibre attaches to the

(1) Centromere

(2) Centrosome

(3) Centriole

(4) Kinetochore

161. How many times does the DNA replicates in one cell cycle?

(1) Once

(2) Twice

(3) Many times

(4) Not at all

162. In meiosis, the chromosome number

(1) Is reduced by half

(2) Increases twice

(3) Sometimes increases, sometimes decreases

(4) None of the above

163. Which of the following statement(s) is/are true?

(1) Cell plate represents the middle lamella between the walls of two adjacent cells.

(2) At the time of cytokinesis, organelles like mitochondria and plastids get distributed between the daughter cells.

(3) Cytokinesis in plant cell is centrifugal and takes place by cell-plate formation while animal cells by furrowing/ cleavage and is centripetal.

(4) All of the above

164. During gamete formation, the enzyme recombinase participates during

(1) Metaphase I

(2) Anaphase II

(3) Prophase I

(4) Prophase II

165. Which of the following types of cell cycle is known as equational division?

(1) Amitosis

(2) Meiosis

(3) Mitosis

(4) 2 and 3

166. Statement A : Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms.

Statement B : Fertilisation restores the diploid phase in the life cycle of sexually reproducing organisms.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

167. The phase of mitosis in which the centriole duplicates in the cytoplasm?

(1) S-phase

(2) G_1-phase

(3) G_2-phase

(4) None of these

168. Which of the following statement(events) is/ are true for mitotic telophase?

(1) Nucleolus, GB and ER reform

(2) Nuclear envelope develops around the chromosome clusters

(3) Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.

(4) All of the above

169. Match Column – I with Column - II.

Column I		Column II	
A	S phase	P	Proteins are synthesized
B	G_2 phase	Q	Inactive phase
C	Quiescent stage	R	Interval between mitosis and initiation of DNA replication
D	G_1 phase	S	DNA replication

(1) A - R, B - Q, C - P, D – S

(2) A - S, B - Q, C - R, D - P

(3) A - S, B - P, C - Q, D – R

(4) A - Q, B - S, C - R, D - P

Title: Unit – 3 (Cell Structure and Function)

170. Assertion : Mitotic division is an equational division.

Reason : Second meiotic division is homotypic.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

171. Synapsis occurs between

(1) Spindle fibres and centromeres.

(2) RNA and ribosomes.

(3) A male and female gamete.

(4) Two homologous chromosomes.

Title: Unit – 3 (Cell Structure and Function)

172. Which of the following statement is incorrect?

(1) Prophase II is simpler than prophase I.

(2) In leptotene stage the chromosomes become gradually visible under light microscope.

(3) Nuclear membrane reappears only in telophase II not in telophase I.

(4) Anaphase II is characterized by the splitting of centromere.

173. In which stage of mitosis, the chromosomes are composed of two chromatids

(1) Prophase and metaphase

(2) Metaphase and telophase

(3) Prophase and telophase

(4) Prophase and anaphase

174. Which of the following is the longest phase of meiosis?

(1) Prophase II

(2) Metaphase I

(3) Prophase I

(4) Metaphase II

175. Statement A : Crossing over is the exchange of genetic material between two non-homologous chromosomes.

Statement B : Crossing over is an enzyme-mediated process and the enzyme involved is called recombinase.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

176. The appearance of recombination nodules on homologous chromosomes during meiosis characterizes:

(1) Bivalent

(2) Sites at which crossing over occurs

(3) Terminalization

(4) Synaptonemal complex

177. What is the function of the enzyme 'recombinase' during meiosis?

(1) Formation of synaptonemal complex

(2) Crossing over between non-sister chromatids

(3) Condensation of chromosomes

(4) Alignment of bivalent chromosomes on equatorial plate

178. Tetrad is made of

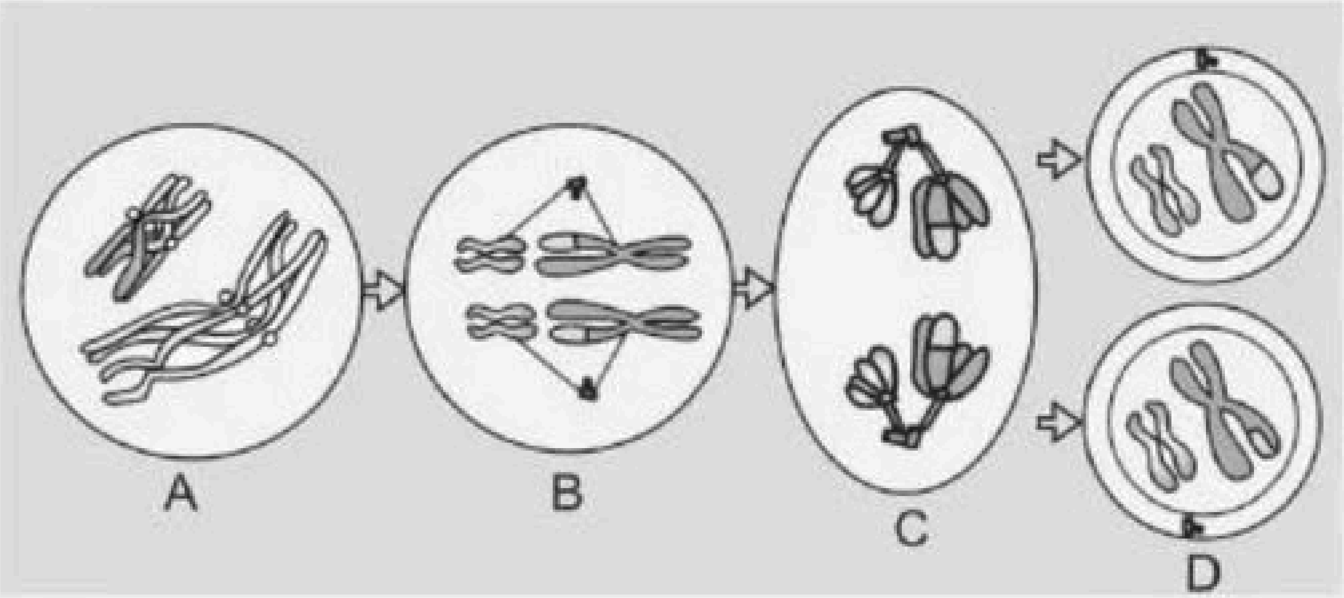
(1) Four homologous chromosomes, each with one chromatids

(2) Four non-homologous chromosomes

(3) Four homologous chromosomes with four chromatids

(4) Two homologous chromosomes, each with two chromatids

179. Name the following stages of meiosis_



(1) A - Early prophase, B - Late prophase,
C - Anaphase, D - Telophase

(2) A - Prophase I, B - Metaphase I,
C - Anaphase II, D - Telophase II

(3) A - Prophase II, B - Metaphase II, C - Anaphase II, D - Telophase II

(4) A - Prophase I, B - Metaphase I, C - Anaphase I, D - Telophase I

180. The complex formed by a pair of synapsed homologous chromosomes is called ________

(1) Bivalent

(2) Axoneme

(3) Equatorial plate

(4) Kinetochore

181. Statement A : The beginning of diplotene is recognised by the dissolution of the synaptonemal complex

Statement B : In oocytes of all vertebrates, diplotene can last for months or years.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

182. Recombination nodule formation is a characteristic of which of the following phases?

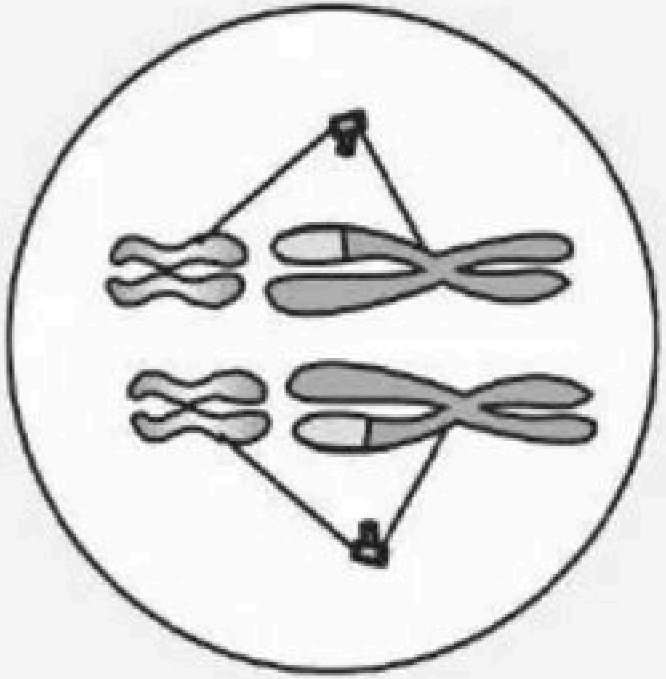
(1) Zygotene

(2) Diplotene

(3) Leptotene

(4) Pachytene

183. Following diagram represents metaphse I, in which ____ are arranged at equator.



(1) Bivalent chromosomes

(2) Tetravalent chromosomes

(3) Monovalent chromosomes

(4) Both 1 and 2

184. The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?

(1) Zygotene

(2) Pachytene

(3) Diplotene

(4) Diakinesis

185. A bivalent of meiosis-I consists of

(1) Two chromatids and one centromere

(2) Two chromatids and two centromere

(3) Four chromatids and two centromere

(4) Four chromatids and four centromere

186. Statement A : During anaphase-I, both homologous chromosomes and sister chromatids separate.

Statement B : During Telophase-I, the nuclear membrane and nucleolus reappear.

(1) Statement-A is correct but Statement-B is incorrect.

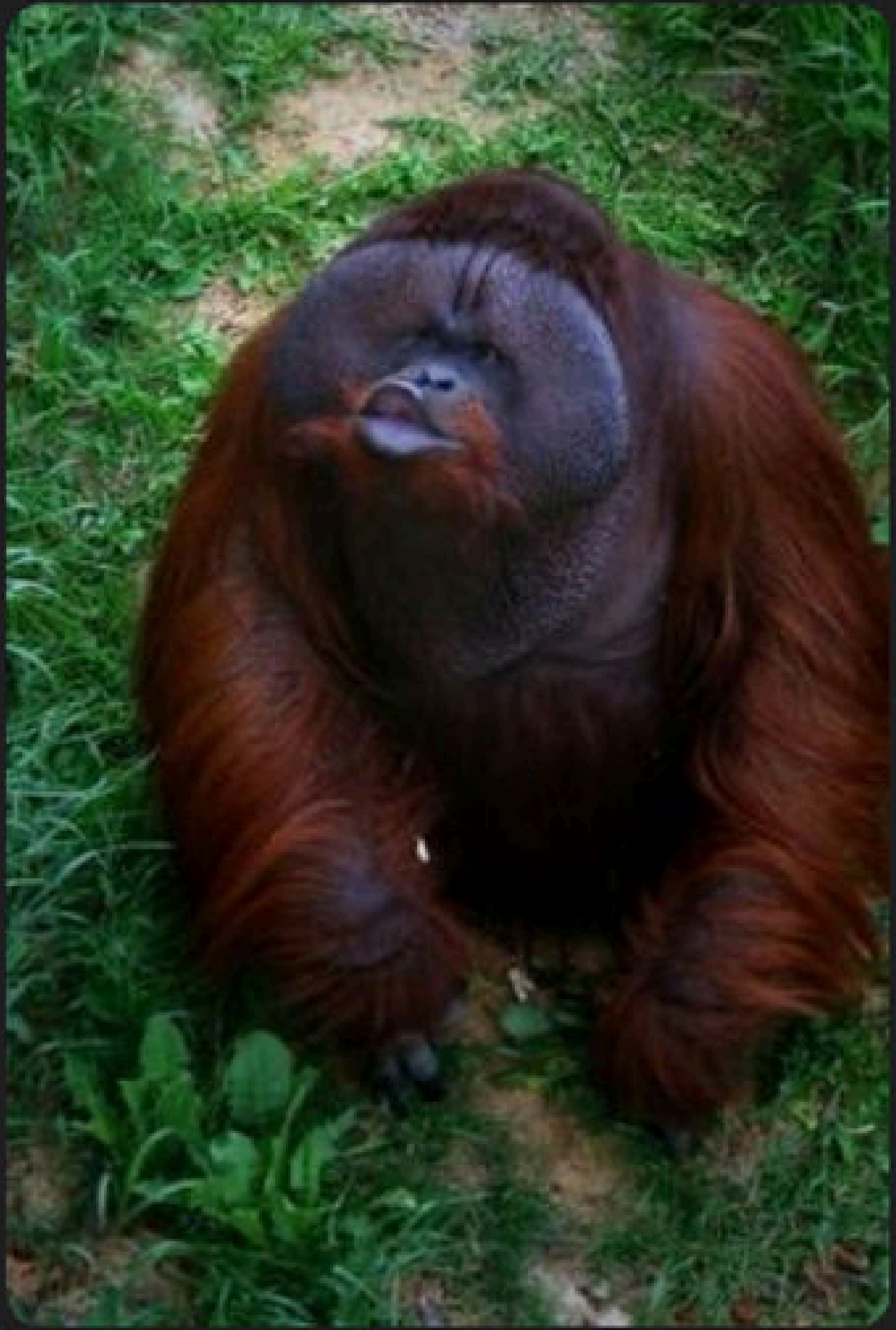
(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

▲ 3 • Asked by Binay Kuma...

Please help me with this doubt



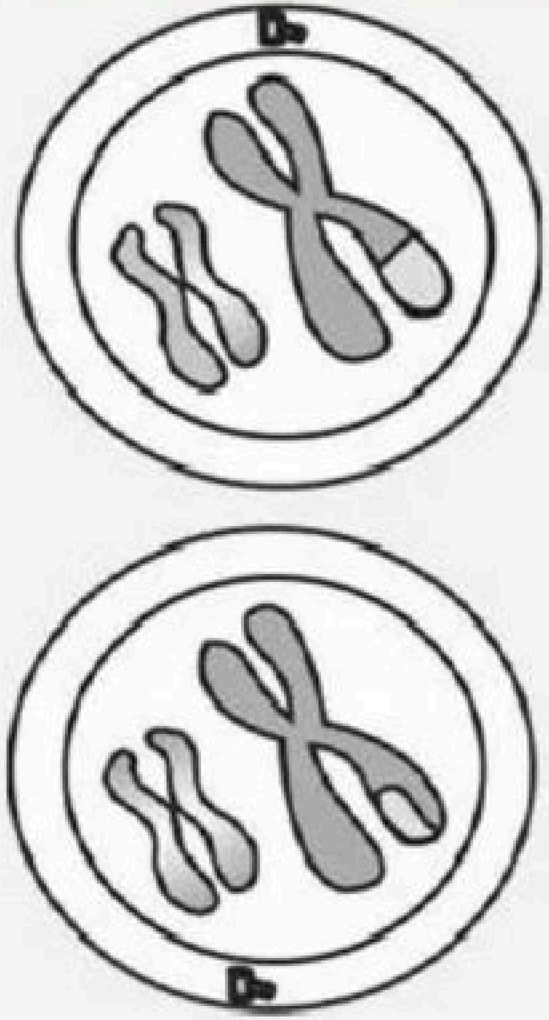
187. In the phase shown by given diagram _

(1) The homologous chromosome separates

(2) Bivalent chromosome separates

(3) Chromosomes start pairing

(4) Cytokinesis follows and this is called dyad of cells



188. The X-shaped structures observed during diplotene are

(1) Chiasmata

(2) Synaptonemal complex

(3) Bivalent complex

(4) 1 and 2

189. In diploid organisms, phenomenon of crossing over is responsible for

(1) Linkages between genes

(2) Recombination and variation

(3) Segregation between genes

(4) Assortment of gene

190. Synaptonemal complex involves

(1) Chromosome pairing

(2) Chromosome movement

(3) Chromosome segregation

(4) Chromosome structure

191. The beginning of diplotene is characterized by

(1) Recombination nodule formation

(2) Synapsis

(3) Dissolution of synaptonemal complex

(4) Crossing over

192. Select the correct statements.

I. Tetrad formation is seen during Leptotene.

II. During Anaphase, the centromeres split and chromatids separate.

III. Terminalization takes place during Pachytene.

IV. Nucleolus, Golgi complex and ER are reformed during Telophase.

V. Crossing over takes place between sister chromatids of homologous chromosome.

Choose the correct answer from the options given below:

(1) I and III only (2) II and IV only

(3) I, III and V only (4) II and V only

Title: Unit – 3 (Cell Structure and Function)

193. Which of the following represents the correct order in prophase I?

(1) Zygotene, diplotene, pachytene, leptotene, diakinesis

(2) Leptotene, zygotene, pachytene, diplotene, diakinesis

(3) Leptotene, diplotene, pachytene, zygotene diakinesis

(4) Pachytene, leptotene, zygotene, diplotene, diakinesis

194. In which stage of meiosis- I, the chromosome is thin, long and thread-like?

(1) Pachytene

(2) Zygotene

(3) Leptotene

(4) Diakinesis

195. The stage during which separation of the paired homologous chromosomes begins is

(1) Diakinesis

(2) Diplotene

(3) Pachytene

(4) Zygotene

196. Statement A : During leptotene stage the chromosomes become gradually visible under the light microscope.

Statement B : During diakinesis stage chromosomes start pairing together and this process of association is called synapsis.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

197. Cross-like configuration when non-sister chromatids of a bivalent comes in contact during the meisosis- I are

(1) Chiasmata

(2) Bivalents

(3) Synapsis

(4) 1 and 3

Title: Unit – 3 (Cell Structure and Function)

198. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Pairing of chromosomes together	P	Bivalent
B	Pair of synapsed homologous chromosomes	Q	Pachytene
C	Appearance of recombination nodules	R	Synapsis
D	Crossing over	S	Recombinase

(1) A-R, B-S, C-Q, D-P (2) A-R, B-S, C-P, D-Q

(3) A-P, B-Q, C-S, D-Q (4) A-R, B-P, C-Q, D-S

199. Statement A : During metaphase-I, the bivalent chromosomes align on the equatorial plate.

Statement B : During anaphase-I, the microtubules from the opposite poles of the spindle attach to the kinetochore of homologous chromosomes.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

100. Statement A : Prophase of the first meiotic division is typically longer and more complex when compared to prophase of mitosis

Statement B : It has been further sub divided into the four sub phases, i.e leptotene, zygotene, pachytene and diplotene.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

THANK YOU

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